



Machine Learning-Based Flood Prediction System

Using Historical Threat Mapping and Live Weather APIs

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PROBLEM STATEMENT

Why today's flood-risk tools fall short



The Limit of Weather Apps

Right now, weather apps show us live data like rain and wind, but they do not tell us if that rain will actually cause a flood in a specific area.



Static Risk Assessment

On the other hand, traditional flood maps are usually drawn once and rarely updated. They cannot react to a sudden, heavy storm.



The Historical Disconnect:

Live weather data and historical disaster records live in separate silos. Current systems struggle to instantly connect a live storm with a location's history of flooding.

Proposed Solution

A single platform connecting live weather to historical flood memory



01

Live Global Weather Search

A platform showing real-time conditions worldwide — search any location to view temperature, rainfall, and wind speed instantly.



02

Location-Based Risk Check

When heavy rainfall is detected in an area, the system automatically looks up that location's flood history.



03

Historical Threat Mapping

EM-DAT international disaster dataset (1988–present) & GPS data from the Dartmouth Flood Observatory (DFO) to check if the area — or nearby zones — has flooded before.



04

AI/ML-Driven Prediction

A trained model fuses live weather with historical vulnerability to output a predicted flood risk, not just raw data.

System Architecture

A five-stage data pipeline, from live signal to risk score



MODEL OPTIONS AT STAGE 4

XGBoost / Random Forest — Best for tabular data — outputs a Low / Medium / High classification.

LSTM Networks — Best for tracking multi-day weather trends — true time-series forecasting.

Beyond Floods

The same pipeline can be extended to predict other hazards

Storms & Cyclones

Dataset: Tropical cyclones, severe weather, blizzards

Predicts: Storm damage from pressure drops & high wind

Why it works: Wind & pressure track cleanly via weather APIs

Landslides

Dataset: Landslides, mudslides, avalanches

Predicts: Slope failure from 48-hr rainfall buildup

Why it works: Sustained rainfall is a strong, trackable trigger

Wildfires

Dataset: Forest & land fires

Predicts: Fire risk score from temp, humidity & wind

Why it works: Just 3 live metrics build a clear risk score